GBODIMOWO ISAAC

Software Engineer & Network Systems Specialist

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kiinnggss.github.io/isaac-portfolio

PROFESSIONAL SUMMARY

Dual-competency engineer bridging full-stack software development with enterprise network routing and hardware-level diagnostics. Experienced in developing scalable TypeScript and Next.js platforms, engineering fault-tolerant Cisco network topologies with sub-second failover (HSRP, Router-on-a-Stick, extended ACLs), and conducting CompTIA A+ certified hardware fault isolation.

EDUCATION & TECHNICAL CREDENTIALS

B.Sc. in Computer Science

Babcock University, Nigeria

Data Structures, Algorithm Design, Relational Database Models, Operating Systems, Computer Networks.

- VERIFIED **CompTIA A+ Certified** (Hardware & Systems Diagnostics Core)
- STUDYING **Cisco CCNA 200-301 Candidate** (Enterprise Routing & Security)
- VERIFIED **New Horizons Technical Certification** (Systems & Networking)

CORE SYSTEMS ENGINEERING

CAR PULL: Peer-to-Peer Transit & Carpooling

DISTRIBUTED WEB ARCHITECTURE

`Next.js, TypeScript, Tailwind CSS, Geospatial Indexing, PostgreSQL, Zustand`

`github.com/kiinnggss/car-pull`

Corridor Indexing: <45ms | **Transit Split Heuristic:** 4-Tier | **Commuter Efficiency:** 3.2x

- **Geospatial Corridor Clustering.** Clustered commuter rides along primary Lagos arterial highways using bounding-box spatial containment with sub-45ms index latency.
- **Automated Route-Split Pricing.** Calculated multi-tier transit cost division based on distance intervals and peak traffic multiplier indices.
- **Driver Operations Cockpit.** Engineered a sequential pickup itinerary with waypoint routing, 5-minute broadcast alerts, and curbside verification.

Enterprise Network Simulation & Defense Topology

ENTERPRISE NETWORKING & SECURITY

`Cisco Packet Tracer, Cisco IOS CLI, CCNA 200-301, HSRP v2, Inter-VLAN (802.1Q), NAT / PAT`

`github.com/kiinnggss/ccna-interactive-hub`

VLAN Segmentation: 100% | **Failover Time:** <3.0s | **Port Protection:** Sticky MAC

- **Hierarchical Topology & Trunking.** Built a 3-tier Core-Distribution-Access hierarchy with 802.1Q trunking and Router-on-a-Stick inter-VLAN routing.
- **High Availability Gateway.** Configured HSRP v2 Active/Standby redundancy between dual Cisco 2911 ISR routers with preemptive failover converging in under 3.0s.
- **Perimeter Defense & WAN NAT.** Enforced extended IPv4 ACLs guarding database server farms, dynamic NAT overload (PAT), and switch port security with sticky MAC learning.

DISTRIBUTED AI & WEB PLATFORMS

Google Antigravity Mobile Controller & Voice Assistant

DISTRIBUTED AI SYSTEMS & WEB ARCHITECTURE

`FastAPI, Python 3.12, Server-Sent Events, Asyncio Subprocesses, Web Speech API, PWA` `github.com/kiinnggss/joyboy_sys`

Stream Latency: <15ms | **Test Coverage:** 66 Tests | **Bundle Size:** 0 KB

- **Process Supervisor & Streaming.** Built an asynchronous FastAPI streaming engine running CLI processes with SSE broadcast and concurrent stderr draining.
- **Zero-Build Voice Interface.** Developed a vanilla PWA frontend with Web Speech dictation, synthesized audio playback, and live terminal monitoring.
- **Zero Trust Tunneling.** Secured endpoints using HMAC SHA-256 session signatures, constant-time PIN authentication, and automated Cloudflare Quick Tunnel provisioning.

Crystal: Mobile Video Downloader & Media Vault

FULL-STACK WEB & MEDIA STREAMING ARCHITECTURE

`FastAPI, React 19, Python 3.12, yt-dlp, Server-Sent Events, Tailwind CSS` `github.com/kiinnggss/video-downloader`

Live Telemetry: <50ms | **Streaming Protocol:** HTTP 206 | **Test Coverage:** 52 Tests

- **Media Stream Extraction & Telemetry.** Built an asynchronous FastAPI pipeline driving yt-dlp workers with sub-50ms Server-Sent Events broadcasting speed, ETA, and progress metrics.
- **Byte-Range Seekable Streaming.** Implemented RFC 7233 HTTP 206 partial content streaming with range and suffix headers enabling instant audio and video playback scrubbing.
- **Encrypted Private Vault & Crystal UI.** Constructed a PIN-protected private vault with PBKDF2/SHA-256 tokens and a mobile-first translucent crystal glassmorphism interface.

TECHNICAL COMPETENCIES MATRIX

Networking & Security	Cisco IOS CLI, VLANs & 802.1Q Trunking, HSRP v2 Redundancy, NAT/PAT, Extended ACLs, IPv4 Subnetting, Cisco Packet Tracer, Switch Port Security.
Software Engineering	TypeScript, JavaScript, Next.js 16 (App Router), React 19, Python 3.12, FastAPI, Tailwind CSS, REST APIs, Server-Sent Events, HTML5, CSS3.
Databases & Tooling	PostgreSQL, SQLite, Cloudflare Quick Tunnels, Git, GitHub, Linux/Unix Shell, Postman, Vitest, Pytest.
Hardware Diagnostics	CompTIA A+ PC Hardware assembly, UEFI/BIOS configuration, power supply and motherboard diagnostics, component fault isolation.

VERIFIED CODE REPOSITORIES & TOPOLOGIES

`kiinnggss/car-pull`
Transit ridesharing architecture with geospatial corridor indexing
`github.com/kiinnggss/car-pull`

`kiinnggss/ccna-interactive-hub`
Enterprise 3-tier Cisco topology with HSRP v2 failover and ACLs
`github.com/kiinnggss/ccna-interactive-hub`

`kiinnggss/joyboy_sys`
Voice controller and streaming supervisor with Cloudflare tunnel
`github.com/kiinnggss/joyboy_sys`

`kiinnggss/video-downloader`
Mobile video inspector, SSE telemetry downloader, and private vault
`github.com/kiinnggss/video-downloader`